Metadata for Abbey St Bathans (ASB), Berwickshire, Scotland, UK

Description of data set	Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. These data consist of measurements of the Earth's natural magnetic field at the remote site (ASBR) and the natural magnetic field plus the field created by GIC at the underline site (ASBU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al (2020) for further details.
Location (in lat/long): underline station (ASBU)	55.848491 N, -2.360845 E
Location (in lat/long): remote station (ASBR)	55.847938 N, -2.365684 E
Distance between DMM sites	308 m
Description of site	Enclosed grazing land on a shallow south facing slope. There was a small wind turbine in another field 400 m away.
Equipment used	Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit)
Installation date	28/08/2018
Removal date	24/05/2019
Number of days collected	267
Data sampling rate	1 Hz
Notes of known data gaps	ASBU: No data from 23-May-2019. ASBR no data from the 23-May-2019.
Qualitative estimate of site quality	Good
Azimuth of HV power line between adjacent pylons relative to true north	-40.35° (i.e. lines run approximately NW-SE)
Alignment of magnetometers	Magnetic north as of 28-Aug-2018
Voltage level of the lines (Torness – Eccles)	400 kV
Height above the ground of each wire (m)	Lower: 12.45 m, Middle: 20.98 m, Upper: 30.29 m
Underline magnetometer	In the centre between both circuits, which is 9 m horizontally from below each set of wires.
File Type	ASCII text (.xyz)
File format	YYYY-MM-DD HH:MM:SS X Y Z Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T). Empty values denoted by: 999999.00
Citation	Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421